GATE ME 90 Days Challenge

Day 15/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Engineering Mathematics	**Calculus**	**94 days to GATE**

How to use this set: 15 questions (7 MCQ, 8 NAT) on limits, continuity, mean value theorems, maxima-minima, definite and improper integrals, multiple integrals. Try them in **40 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

The value of $\lim_{x \to 0} (e^{x} - e^{-x} - 2x) / (x - \sin x)$ is

(A) 1 **(B)** 1/2

(C) 2 **(D)** 0

Q2 NAT

The value of $\lim_{x \to \infty} (1 + 3/x)^{2x}$ (round off to two decimal places) is _____

Q3 MCQ

For $f(x) = x|x|$, which statement is correct at $x = 0$?

(A) f is continuous but not differentiable **(B)** f is differentiable and $f'(0) = 0$

(C) f is not continuous **(D)** f is differentiable and $f'(0) = 1$

Q4 NAT

The value of c given by Lagrange's mean value theorem for $f(x) = \sqrt{x}$ on the interval [4, 9] is _____ (round off to two decimal places)

Q5 MCQ

For $f(x) = \sin x + \cos x$ on $[0, \pi/2]$, the value of c given by Rolle's theorem is

(A) π/6 **(B)** π/3

(C) π/4 **(D)** π/2

Q6 NAT

The absolute maximum value of $f(x) = x^{3} - 6x^{2} + 9x + 15$ on the closed interval [0, 5] is _____

Q7 MCQ

The minimum value of $f(x, y) = x^4 + y^4 - 4xy$ is

(A) 0
(B) -1
(C) 2
(D) -2

Q8 NAT

The minimum value of $x^2 + y^2$ subject to the constraint $x + 2y = 10$ is _____

Q9 MCQ

The value of $\int_0^{\pi/2} \sqrt{(\sin x)} / (\sqrt{(\sin x)} + \sqrt{(\cos x)})\, dx$ is

(A) $\pi/2$
(B) $\pi/4$
(C) 1
(D) 0

Q10 NAT

The value of $\int_0^{\pi} x \sin^2 x\, dx$ is _____ (round off to two decimal places)

Q11 MCQ

The value of the improper integral $\int_{-\infty}^{\infty} dx / (x^2 + 2x + 5)$ is

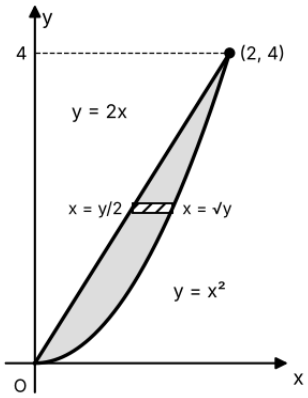
(A) $\pi/2$
(B) π
(C) $\pi/4$
(D) 2π

Q12 NAT

The value of $\int_0^{\infty} x^4 e^{-x^2}\, dx$ is _____ (round off to three decimal places)

Q13 MCQ

The region of integration of $\int_0^4 \int_{y/2}^{\sqrt{y}} f(x, y)\, dx\, dy$ is shown in the figure. On reversing the order of integration, the integral becomes



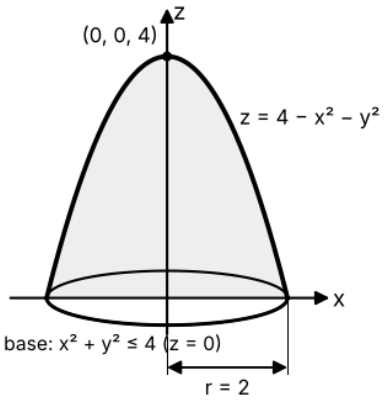
(A) $\int_0^2 \int_{2x}^{x^2} f(x, y)\, dy\, dx$
(B) $\int_0^4 \int_{x^2}^{2x} f(x, y)\, dy\, dx$
(C) $\int_0^2 \int_{x/2}^{\sqrt{x}} f(x, y)\, dy\, dx$
(D) $\int_0^2 \int_{x^2}^{2x} f(x, y)\, dy\, dx$

Q14 NAT

The value of $\iint_R e^{-(x^2 + y^2)}\, dx\, dy$, where R is the disc $x^2 + y^2 \le 1$, is _____ (round off to three decimal places)

Q15 NAT

The volume of the solid bounded by the paraboloid $z = 4 - x^2 - y^2$ and the plane $z = 0$ (shown) is _____ (round off to two decimal places)



Answer Key

Q	Type	Answer
Q1	MCQ	(C) 2
Q2	NAT	403.43 (accept 401 to 406)
Q3	MCQ	(B) f is differentiable and f′(0) = 0
Q4	NAT	6.25
Q5	MCQ	(C) π/4
Q6	NAT	35
Q7	MCQ	(D) −2
Q8	NAT	20
Q9	MCQ	(B) π/4
Q10	NAT	2.47 (accept 2.45 to 2.49)
Q11	MCQ	(A) π/2
Q12	NAT	0.665 (accept 0.660 to 0.670)
Q13	MCQ	(D) $\int_0^2 \int_{x^2}^{2x} f(x, y)\, dy\, dx$
Q14	NAT	1.986 (accept 1.976 to 1.996)
Q15	NAT	25.13 (accept 25.0 to 25.3)

Detailed Solutions

Q1 **Answer: (C) 2**

Concept: For a 0/0 limit, expand numerator and denominator in Maclaurin series and compare the lowest powers.

$e^x - e^{-x} = 2 \sinh x = 2x + x^3/3 + \ldots$, so the numerator = $\mathbf{x^3/3} + O(x^5)$.

$x - \sin x = \mathbf{x^3/6} + O(x^5)$.

Limit = (1/3) ÷ (1/6) = **2**

Exam tip: L'Hospital needs three rounds here; the series method settles it in two lines. Keep e^x, sin x, cos x, ln(1+x) series to the x^3 term at your fingertips.

Q2 **Answer: 403.43 (accept 401 to 406)**

Concept: 1^∞ form: $\lim f^g = e^L$, where $L = \lim g \cdot (f - 1)$.

Here $f - 1 = 3/x$ and $g = 2x$, so $L = \lim 2x \cdot (3/x) = \mathbf{6}$.

Limit = $e^6 = \mathbf{403.43}$

Exam tip: Remember $\lim (1 + a/x)^{bx} = e^{ab}$. Writing the answer as 1 (because 1 raised to anything is 1) is the classic trap.

Q3 **Answer: (B) f is differentiable and f′(0) = 0**

Concept: Test differentiability with the left and right difference quotients.

$f(x) = x^2$ for $x \geq 0$ and $-x^2$ for $x < 0$, so $f(0) = 0$ and f is continuous.

Right derivative: $\lim_{h \to 0+} h \cdot h / h = 0$. Left derivative: $\lim_{h \to 0-} h(-h)/h = 0$.

Both equal 0, so f is **differentiable with f′(0) = 0**. (In fact $f'(x) = 2|x|$, which is not differentiable at 0, so f″(0) does not exist.)

Exam tip: |x| has a corner at 0, but multiplying by x flattens it. Do not mark every function containing |x| as non-differentiable.

Q4 **Answer: 6.25**

Concept: Lagrange's MVT: $f'(c) = [f(b) - f(a)] / (b - a)$ for some c in (a, b).

$f'(x) = 1/(2\sqrt{x})$. Chord slope $= (\sqrt{9} - \sqrt{4})/(9 - 4) = (3 - 2)/5 =$ **1/5**.

$1/(2\sqrt{c}) = 1/5 \rightarrow \sqrt{c} = 5/2 \rightarrow c =$ **6.25**, which lies in (4, 9) ✓

Exam tip: Always check that c lands inside the open interval; if it does not, re-check the algebra.

Q5 **Answer: (C) π/4**

Concept: Rolle's theorem: f continuous on [a, b], differentiable on (a, b) and $f(a) = f(b) \Rightarrow f'(c) = 0$ for some c in (a, b).

$f(0) = 0 + 1 = 1$ and $f(\pi/2) = 1 + 0 = 1$, so $f(a) = f(b)$ ✓

$f'(x) = \cos x - \sin x = 0 \rightarrow \tan c = 1 \rightarrow c =$ **π/4**

Exam tip: Verify f(a) = f(b) first. π/3 and π/6 are tempting but give $f'(c) \neq 0$.

Q6 **Answer: 35**

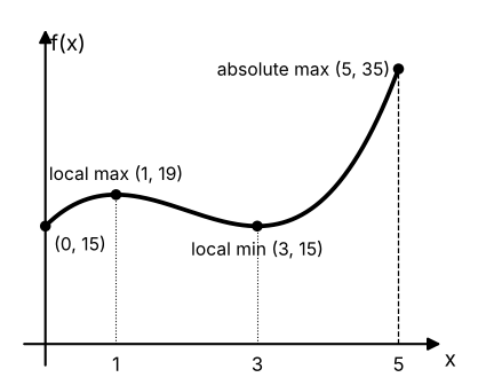
Concept: On a closed interval, the absolute maximum is the largest of f at the critical points and at the two end points.

$f'(x) = 3x^2 - 12x + 9 = 3(x - 1)(x - 3) = 0 \rightarrow x = 1, 3$.

$f(0) = 15$, $f(1) = 1 - 6 + 9 + 15 =$ **19** (local max), $f(3) = 27 - 54 + 27 + 15 = 15$ (local min), $f(5) = 125 - 150 + 45 + 15 =$ **35**.

Absolute maximum = **35**, at the end point x = 5 (figure).

Exam tip: 19 is only the local maximum. A local maximum need not be the absolute maximum; always evaluate the end points.



Q7 **Answer: (D) −2**

Concept: Stationary points from $f_x = f_y = 0$; classify with $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$: $rt - s^2 > 0$ and $r > 0 \rightarrow$ minimum.

$f_x = 4x^3 - 4y = 0 \rightarrow y = x^3$; $f_y = 4y^3 - 4x = 0 \rightarrow x^9 = x \rightarrow x = 0, \pm 1$.

Stationary points: (0, 0), (1, 1), (−1, −1).

At (0, 0): $r = t = 0$, $s = -4 \rightarrow rt - s^2 = -16 < 0 \rightarrow$ saddle point.

At (±1, ±1): $r = 12x^2 = 12$, $t = 12$, $s = -4 \rightarrow rt - s^2 = 128 > 0$, $r > 0 \rightarrow$ minimum, $f = 1 + 1 - 4 =$ **−2**

Exam tip: f(0, 0) = 0 is not the minimum: (0, 0) is a saddle. Always run the $rt - s^2$ test.

Q8 **Answer: 20**

Concept: Lagrange multipliers: $\nabla f = \lambda \nabla g$ at the constrained optimum.

$2x = \lambda$ and $2y = 2\lambda \rightarrow y = 2x$.

Substitute in $x + 2y = 10$: $5x = 10 \rightarrow x = 2$, $y = 4$.

Minimum = $2^2 + 4^2 =$ **20**

Exam tip: Check geometrically: it is the square of the distance from the origin to the line, $(10/\sqrt{5})^2 = 20$.

Q9 **Answer: (B) π/4**

Concept: King's property: $\int_0^a f(x)\,dx = \int_0^a f(a - x)\,dx$.

Replacing x by π/2 − x swaps sin and cos: $I = \int_0^{\pi/2} \sqrt{\cos x} / (\sqrt{\cos x} + \sqrt{\sin x})\,dx$.

Adding the two forms: $2I = \int_0^{\pi/2} 1\,dx = \pi/2$.

$I =$ **π/4**

Exam tip: Whenever sin and cos appear symmetrically over [0, π/2], the answer is half the interval length.

Q10 **Answer: 2.47 (accept 2.45 to 2.49)**

Concept: $\int_0^a x\,f(x)\,dx = (a/2)\int_0^a f(x)\,dx$ when $f(a - x) = f(x)$.

$\sin^2(\pi - x) = \sin^2 x$, so $I = (\pi/2)\int_0^{\pi} \sin^2 x\,dx$.

$\int_0^{\pi} \sin^2 x\,dx = \pi/2$ (average value ½ over a full period).

$I = (\pi/2)(\pi/2) = \pi^2/4 =$ **2.47**

Exam tip: This property removes the x factor without integration by parts.

Q11 **Answer: (A) π/2**

Concept: Complete the square and use $\int dx/(u^2 + a^2) = (1/a)\tan^{-1}(u/a)$.

$x^2 + 2x + 5 = (x + 1)^2 + 2^2$.

$I = (1/2)[\tan^{-1}((x + 1)/2)]_{-\infty}^{\infty} = (1/2)(\pi/2 + \pi/2) =$ **π/2**

The integrand behaves like $1/x^2$ at both ends, so the integral converges.

Exam tip: Over (−∞, ∞) the shift x + 1 does not change the limits; only a (here 2) matters: answer = π/a.

Q12 **Answer: 0.665 (accept 0.660 to 0.670)**

Concept: Gamma function: $\Gamma(n) = \int_0^{\infty} t^{n-1} e^{-t}\, dt$, $\Gamma(n+1) = n\Gamma(n)$, $\Gamma(\tfrac{1}{2}) = \sqrt{\pi}$.

Put $t = x^2$, $x = t^{1/2}$, $dx = dt / (2t^{1/2})$: $I = \tfrac{1}{2} \int_0^{\infty} t^{3/2} e^{-t}\, dt = \tfrac{1}{2}\, \Gamma(5/2)$.

$\Gamma(5/2) = (3/2)(1/2)\Gamma(1/2) = (3/4)\sqrt{\pi}$.

$I = (3/8)\sqrt{\pi} = 0.375 \times 1.7725 =$ **0.665**

Exam tip: General result: $\int_0^{\infty} x^m e^{-x^2}\, dx = \tfrac{1}{2}\, \Gamma((m+1)/2)$.

Q13 **Answer: (D) $\int_0^2 \int_{x^2}^{2x} f(x, y)\, dy\, dx$**

Concept: Sketch the region from the given limits, then read the new limits from a strip drawn in the other direction.

Inner limits $x = y/2$ (the line $y = 2x$) and $x = \sqrt{y}$ (the parabola $y = x^2$); they meet at (0, 0) and (2, 4).

For a vertical strip at x ($0 \le x \le 2$): y runs from the parabola $y = x^2$ (lower) to the line $y = 2x$ (upper).

New integral = $\int_0^2 \int_{x^2}^{2x} f\, dy\, dx$. Check with $f = 1$: area = $\int_0^2 (2x - x^2)\, dx = 4/3$ by both orders.

Exam tip: Option (A) swaps the curves (it gives $-4/3$ for $f = 1$). Test any reversal with $f = 1$: both orders must give the same positive area.

Q14 **Answer: 1.986 (accept 1.976 to 1.996)**

Concept: For circular regions, switch to polar coordinates: $x^2 + y^2 = r^2$, $dx\, dy = r\, dr\, d\theta$.

$I = \int_0^{2\pi} \int_0^1 e^{-r^2}\, r\, dr\, d\theta$.

Inner: $\int_0^1 r\, e^{-r^2}\, dr = \tfrac{1}{2}(1 - e^{-1}) = 0.3161$.

$I = 2\pi \times 0.3161 = \pi(1 - 1/e) =$ **1.986**

Exam tip: The Jacobian r is what makes e^{-r^2} integrable. Over the whole plane the same steps give π, the basis of $\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}$.

Q15 **Answer: 25.13 (accept 25.0 to 25.3)**

Concept: Volume = $\iint$ (height) dA over the base region; use polar coordinates for a circular base.

The paraboloid meets $z = 0$ on the circle $x^2 + y^2 = 4$, so the base is a disc of radius 2 and height $= 4 - r^2$.

$V = \int_0^{2\pi} \int_0^2 (4 - r^2)\, r\, dr\, d\theta = 2\pi\, [2r^2 - r^4/4]_0^2 = 2\pi(8 - 4) = 8\pi$

$V = 8\pi =$ **25.13**

Exam tip: A paraboloid cap is exactly half of its circumscribing cylinder: $\tfrac{1}{2} \times \pi(2)^2 \times 4 = 8\pi$. A quick check worth remembering.

What next?

Day 15 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 16/90: Theory of Machines, Dynamics of Linkages & Flywheels. Comment **GATE90** on our reel to get all links.
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